

$$\begin{aligned}
 349. \quad \frac{x^3 - y^3}{x^2 - y^2} - \frac{3xy}{x + y} &= \frac{(x - y)(x^2 + y^2 + xy)}{(x - y)(x + y)} - \frac{3xy}{(x + y)} \\
 &= \frac{(x^2 + y^2 + xy) - 3xy}{(x + y)} = \frac{(x^2 + y^2 - 2xy)}{(x + y)} \\
 &= \frac{(x - y)^2}{(x + y)} = \frac{(5 - 3)^2}{(5 + 3)} = \frac{4}{8} = \frac{1}{2}.
 \end{aligned}$$

$$\begin{aligned}
 350. \quad (a + b + c) &= 11 \Rightarrow (a + b + c)^2 = (11)^2 = 121 \\
 &\Rightarrow a^2 + b^2 + c^2 + 2(ab + bc + ca) = 121 \\
 &\Rightarrow a^2 + b^2 + c^2 + 2 \times 20 = 121 \\
 &\Rightarrow a^2 + b^2 + c^2 = 121 - 40 = 81. \\
 \therefore a^3 + b^3 + c^3 - 3abc &= (a + b + c) \\
 &\quad (a^2 + b^2 + c^2 - ab - bc - ca) \\
 &= 11(81 - 20) = 11 \times 61 = 671.
 \end{aligned}$$

$$\begin{aligned}
 351. \quad \text{Since } (x - y) + (y - z) + (z - x) &= 0, \\
 \text{so } (x - y)^3 + (y - z)^3 + (z - x)^3 &= 3(x - y)(y - z)(z - x). \\
 \text{Since } (x^2 - y^2) + (y^2 - z^2) + (z^2 - x^2) &= 0, \\
 \text{so } (x^2 - y^2)^3 + (y^2 - z^2)^3 + (z^2 - x^2)^3 &= 3(x^2 - y^2)(y^2 - z^2)(z^2 - x^2). \\
 \therefore \text{Given exp.} &= \frac{3(x - y)(y - z)(z - x)}{3(x^2 - y^2)(y^2 - z^2)(z^2 - x^2)} \\
 &= \frac{1}{(x + y)(y + z)(z + x)} \\
 &= [(x + y)(y + z)(z + x)]^{-1}.
 \end{aligned}$$

$$\begin{aligned}
 352. \quad \frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab} &= \frac{a^3 + b^3 + c^3}{abc} = \frac{3abc}{abc} = 3. \\
 [\because a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 &= 3abc] \\
 353. \quad a^3 + b^3 + c^3 - 3abc &= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) \\
 &= (29 + 24 + 27)[(29)^2 + (24)^2 + (27)^2 \\
 &\quad - 29 \times 24 - 24 \times 27 - 27 \times 29] \\
 &= 80(841 + 576 + 729 - 696 - 648 - 783) = 80 \times 19 = 1520.
 \end{aligned}$$

$$\begin{aligned}
 354. \quad \frac{x^2 - 1}{x} = 4 &\Rightarrow \left(\frac{x^2 - 1}{x}\right)^3 = 4^3 = 64 \\
 &\Rightarrow \frac{x^6 - 1 - 3x^2(x^2 - 1)}{x^3} = 64 \\
 &\Rightarrow \frac{x^6 - 1 - 3x^2 \cdot 4x}{x^3} = 64 \\
 &\Rightarrow \frac{x^6 - 1 - 12x^3}{x^3} = 64x^3 \\
 &\Rightarrow x^6 - 1 = 76x^3 \Rightarrow \frac{x^6 - 1}{x^3} = 76.
 \end{aligned}$$

$$\begin{aligned}
 355. \quad \text{Let } a = 32, b = 79, c = -111. \text{ Then, } a + b + c &= 0. \\
 \text{So, } a^3 + b^3 + c^3 = 3abc &\Rightarrow (32)^3 + (79)^3 - (111)^3 \\
 &= 3 \times 32 \times 79 \times (-111) \\
 \therefore \text{Given exp.} &= -(3 \times 32 \times 79 \times 111) + 3 \times 32 \times 79 \\
 &\quad \times 111 = 0.
 \end{aligned}$$

$$\begin{aligned}
 356. \quad (x + y) = 3 &\Rightarrow (x + y)^2 = 9 \\
 &\Rightarrow x^2 + y^2 + 2xy = 9 \\
 &\Rightarrow x^2 + y^2 = 9 - 2xy = 9 - 4 = 5. \\
 \therefore x^3 + y^3 &= (x + y)(x^2 + y^2 - xy) = 3(5 - 2) \\
 &= 3 \times 3 = 9. [\because x + y = 3, x^2 + y^2 = 5, xy = 2]
 \end{aligned}$$

$$\begin{aligned}
 357. \quad \text{Given expression} &= \frac{(x + y - z)(x - y + z)}{(x + z + y)(x + z - y)} \\
 &\quad + \frac{(y + x - z)(y - x + z)}{(x + y + z)(x + y - z)} + \frac{(z + x - y)(z - x + y)}{(y + z + x)(y + z - x)} \\
 &= \frac{(x + y - z)}{(x + y + z)} + \frac{(y - x + z)}{(x + y + z)} + \frac{(x - y + z)}{(x + y + z)} \\
 &= \frac{(x + y - z) + (y - x + z) + (x - y + z)}{(x + y + z)} \\
 &= \frac{x + y + z}{x + y + z} = 1.
 \end{aligned}$$

$$358. \quad \frac{a}{p} + \frac{b}{q} + \frac{c}{r} = 0 \Rightarrow aqr + bpr + cpq = 0 \quad \dots(i)$$

$$\begin{aligned}
 \frac{p}{a} + \frac{q}{b} + \frac{r}{c} &= 1 \Rightarrow \left(\frac{p}{a} + \frac{q}{b} + \frac{r}{c}\right)^2 = 1 \\
 \Rightarrow \frac{p^2}{a^2} + \frac{q^2}{b^2} + \frac{r^2}{c^2} + 2\left(\frac{pq}{ab} + \frac{pr}{ac} + \frac{qr}{bc}\right) &= 1 \\
 \Rightarrow \frac{p^2}{a^2} + \frac{q^2}{b^2} + \frac{r^2}{c^2} + \frac{2(pqc + prb + qra)}{abc} &= 1 \\
 \Rightarrow \frac{p^2}{a^2} + \frac{q^2}{b^2} + \frac{r^2}{c^2} &= 1 \quad [\text{Using (i)}]
 \end{aligned}$$

$$\begin{aligned}
 359. \quad x^4 - y^4 = 15 &\Rightarrow (x^2 - y^2)(x^2 + y^2) = 15 = 1 \times 15 \text{ or } 3 \times 5 \\
 &\Rightarrow (x^2 - y^2) = 3 \text{ and } (x^2 + y^2) = 5 \\
 &\quad [\because x \text{ and } y \text{ are natural numbers}] \\
 &\Rightarrow 2x^2 = 8 \Rightarrow x^2 = 4 \Rightarrow x = 2. \\
 &\quad [\text{Adding above two equations}] \\
 \text{Putting } x = 2 \text{ in } x^2 + y^2 = 5, \text{ we get: } y &= 1. \\
 \therefore (x^4 + y^4) &= (2^4 + 1^4) = (16 + 1) = 17.
 \end{aligned}$$

$$360. \quad 26 \text{ trees have 25 gaps between them. Hence, required distance} = \left(\frac{225}{25}\right) \text{ m} = 9 \text{ m}.$$

$$\begin{aligned}
 361. \quad \text{Each row contains 12 plants.} \\
 12 \text{ plants have 11 gaps between them. So, total length of} \\
 \text{gaps} = (11 \times 2) \text{ metres and 1 metre is left on each side.} \\
 \therefore \text{Length} = (22 + 2) \text{ m} = 24 \text{ m}.
 \end{aligned}$$

$$\begin{aligned}
 362. \quad \text{Let the length of the side be } x \text{ metres.} \\
 \text{Then, number of posts required when placed 6 m apart} \\
 = \left(\frac{x}{6} + 1\right).
 \end{aligned}$$

$$\begin{aligned}
 \text{And, number of posts required when placed 8 m apart} \\
 = \left(\frac{x}{8} + 1\right).
 \end{aligned}$$

$$\therefore \left(\frac{x}{6} + 1\right) - 5 = \left(\frac{x}{8} + 1\right) \Leftrightarrow \frac{x}{6} - \frac{x}{8} = 5 \Leftrightarrow \frac{x}{24} = 5 \Leftrightarrow x = 120 \text{ m}.$$

$$\text{So, number of posts bought} = \left(\frac{120}{8} + 1\right) = 16.$$

$$363. \quad \text{Gap between 2 consecutive trees} = \left(\frac{2000}{200}\right) \text{ m} = 10 \text{ m}.$$

$$[\because 201 \text{ trees have 200 gaps between them}]$$

$$\begin{aligned}
 \therefore \text{Required number of trees} &= \left(\frac{50 \times 1000}{10} + 1\right) \\
 &= (5000 + 1) = 5001.
 \end{aligned}$$